The Effect of the Bologna Reform on the Graduate Labour Market

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Motivation

- Rare causal estimates of returns to higher education for low- and middle-income countries
- Low– and middle–income countries might have distinct and heterogeneous features of their institutions, such as labour markets and universities
- No paper that evaluates the results of the Bologna reform in Russia

Research question

- This paper estimates the short–term causal effect of one year of university education on wages and employment in Russia

Literature

- Causal estimates of returns to higher and vocational education
  - Returns to an additional year are positive and significant
    (Webbink, 2007; Maurin and McNally, 2008; Suhonen and Karhunen, 2019)
  - Changing the length of education has no effect on labour outcomes
    (Oosterbeek and Webbink, 2007; Pischke and von Wachter, 2008)
- Few papers on low– and middle–income countries
  (Arteaga, 2018; Dai et al., 2021)

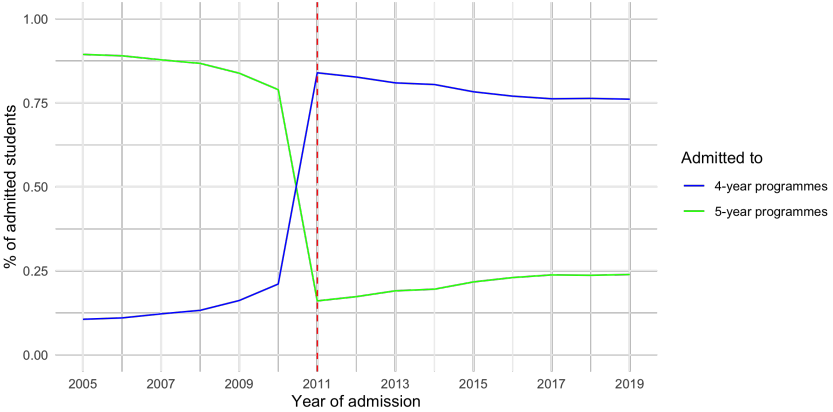
Literature

- Bologna reform in European countries
  - Positive results of the reform on wages in Germany and Portugal and on employment in Italy (Bosio and Leonardi, 2011; Garra, 2013; Neugebauera and Weiss, 2018)
  - Negative impact on wages in Italy and on employment in Germany and Slovenia (Bosio and Leonardi, 2011; Farcnik and Domadenik, 2012; Neugebauera and Weiss, 2018)
- No paper about the results of the Bologna reform in Russia

Institutional background

- 1992: 4-year programmes were optional parallel to the 5-year degrees
- 2003: Russia signed the Bologna Declaration
- 2011: completed transformation from 5-year to 4-year programmes

Figure: The share of 4– and 5–year full–time admitted students in public and private universities in 2005–2019



Notes: The red dashed line shows the year of the implementation of the reform.

Data

- National Survey of Graduate Employment
- Repeated cross-sectional sample of 2010–2015 graduates
- 4– and 5–year university graduates
- 20–30 years old graduates who are not currently students
- Only full-time programmes
- In total, there are 11,332 observations Descriptive statistics

Empirical strategy

$$Y_{ipry} = \gamma_0 + \gamma_1 Treated_p + \gamma_2 Treated_p * Post_y + \mathbf{X_{ipr}} + \lambda_p + \mu_y + \eta_r + \varepsilon_{ipry}$$

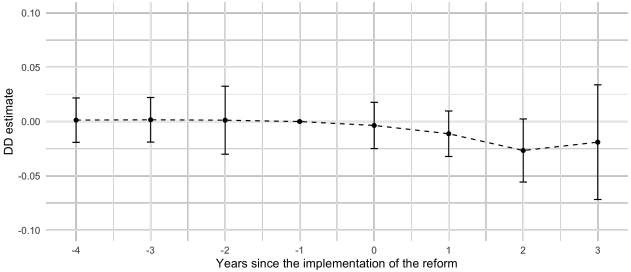
Y_{ipr} is the labour market outcome of graduate i from programme p in region r
$Treated_p$ is a dummy variable equal to 1 if a student has graduated from a 4–year programme p and 0 otherwise
$Treated_p * Post_y$ is the interaction of $Treated_p$ and a dummy variable for the year of graduation 2015
$\mathbf{X_{ipr}}$ are exogenous individual characteristics (gender, type of institution (public/private), and funding (state–funded/fee–based))
λ_p are programme fixed effects
η_r are region fixed effects
μ_y are the year of graduation fixed effects.

Identification strategy

- Bologna reform offers a source of exogenous variation in the length of university education
- University applicants could not affect their decision to start 4– or 5–year programmes
- The treatment is orthogonal to the potential outcomes, such as ability or self-selection

Test for parallel pre–reform trends

Figure: Event-study estimates of a one–year reduction of higher education on employment



Notes: The figure shows the regression coefficients of interaction between the treatment variable and time period dummies with 95% confidence intervals. The omitted period is -1, the year before the implementation of the reform.

Results

Table: The OLS estimates of the association between years of higher education and log monthly wages and employment

	Log monthly wages			*Employed*		
	All	Male	Female	All	Male	Female
	(1)	(2)	(3)	(4)	(5)	(6)
Years of education	0.045**	0.035*	0.057**	0.023**	0.015	0.027**
	(0.013)	(0.014)	(0.017)	(0.007)	(0.010)	(0.011)
Pre–reform mean	10.02	10.02	10.02	0.88	0.88	0.88
Observations	6,207	2,771	3,436	11,332	4,783	6,549

Notes: Robust standard errors clustered at the programme and the graduation year level in parentheses.
*p<0.1; **p<0.05; ***p<0.01

Results

Table: The effect of a one–year reduction of higher education on log monthly wages, hourly wages, and employment for all

	Log monthly wages		*Log hourly wages*		*Employed*	
	(1)	(2)	(3)	(4)	(5)	(6)
Treated*Post	−0.010	−0.018	−0.021	−0.026	−0.014	−0.015*
	(0.015)	(0.012)	(0.016)	(0.022)	(0.007)	(0.007)
Year and region FE	✓	✓	✓	✓	✓	✓
Ind. controls		✓		✓		✓
Pre–reform mean	10.02	10.02	4.91	4.91	0.88	0.88
Observations	6,207	6,207	6,207	6,207	11,332	11,332

Notes: Robust standard errors clustered at the programme and the graduation year level in parentheses.
$^{*}p<0.1$; $^{**}p<0.05$; $^{***}p<0.01$

Results

Table: The effect of a one–year reduction of higher education on log monthly wages and employment separately by gender

	Log monthly wages		*Employed*	
	Male	Female	Male	Female
	(1)	(2)	(3)	(4)
Treated*Post	-0.046^{**}	0.017	-0.020	-0.012
	(0.015)	(0.015)	(0.032)	(0.022)
Pre–reform mean	10.13	9.92	0.95	0.82
Observations	2,771	3,436	4,783	6,549

Notes: Robust standard errors clustered at the programme and the graduation year level in parentheses.

*p<0.1; **p<0.05; ***p<0.01

Results

Table: The effect of a one–year reduction of higher education on log monthly wages and employment separately by funding

	Log monthly wages		*Employed*	
	State–funded	Fee–based	State–funded	Fee–based
	(1)	(2)	(3)	(4)
Treated*Post	−0.013	−0.001	−0.001	−0.021
	(0.020)	(0.010)	(0.015)	(0.015)
Pre–reform mean	10.02	10.01	0.88	0.86
Observations	3,730	2,477	6,697	4,635

Notes: Robust standard errors clustered at the programme and the graduation year level in parentheses.

*p<0.1; **p<0.05; ***p<0.01

Mechanism

- As human capital theory suggests, shortening the length of university study period reduces the direct and indirect costs of obtaining a degree
  - the reduction in the length of study leads to the decreased productivity of a graduate, which will result in lower wages and unfavourable labour market opportunities
  - some students can compensate the reduction of the duration of their studies and invest in the accumulation of the human capital by studying more outside of the classroom

Mechanism

Table: The effect of a one–year reduction of higher education on combining study and work and the level of on-the-job training

	Combining study and work		*On-the-job training*	
	Male	Female	Male	Female
	(1)	(2)	(3)	(4)
Treated*Post	0.081^{***}	-0.042^{**}	0.072^{**}	-0.045^{**}
	(0.015)	(0.012)	(0.024)	(0.013)
Pre–reform mean	0.28	0.20	0.14	0.13
Observations	4,783	6,549	1,385	1,747

Notes: Robust standard errors clustered at the programme and the graduation year level in parentheses.

$^{*}p<0.1$; $^{**}p<0.05$; $^{***}p<0.01$

Possible threats to the identification strategy

- Changes in the cohort composition Share of applicants Number of applicants Characteristics of students Share of graduates
- An introduction of the standardised exam for admission to universities Characteristics of applicants
- The 2015 Russian financial crisis Employment in 2014–2018

Conclusion

- The Bologna reform offers exogenous variation in the length of university education
- No adverse effect of a 20% reduction in the length of higher education on employment
- No adverse effects on wages for graduates who spent more time studying
  - there are negative effects for students who worked during their studies
- I provide suggestive evidence for the compensation–based mechanism to explain these results
- Long-term effects are to be estimated

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Table: Summary statistics

Variable	Observations	Mean	SD	Min	Max
4–year university graduates					
Male	1,722	0.41	0.49	0	1
Public institution	1,722	0.94	0.23	0	1
State–funded	1,722	0.53	0.50	0	1
Employed	1,722	0.82	0.38	0	1
Monthly wage, RUB	887	23,279	11,092	5,000	130,000
5–year university graduates					
Male	9,610	0.42	0.49	0	1
Public institution	9,610	0.95	0.21	0	1
State–funded	9,610	0.60	0.49	0	1
Employed	9,610	0.86	0.35	0	1
Monthly wage, RUB	5,320	24,539	12,009	5,000	120,000

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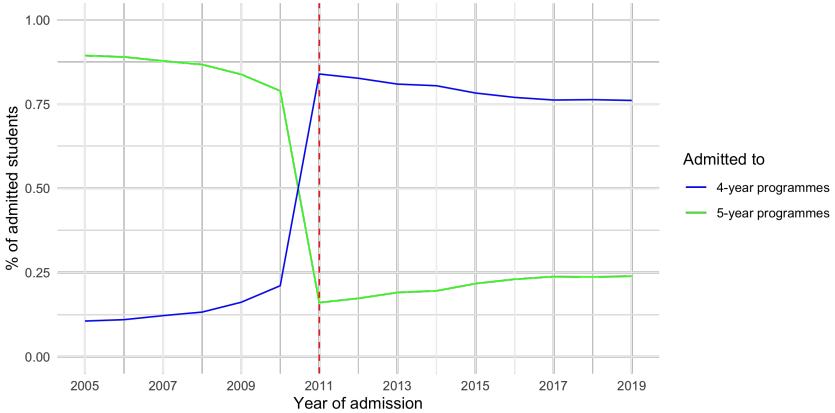
Figure: The share of 4– and 5–year full–time admitted students in public and private universities in 2005–2019



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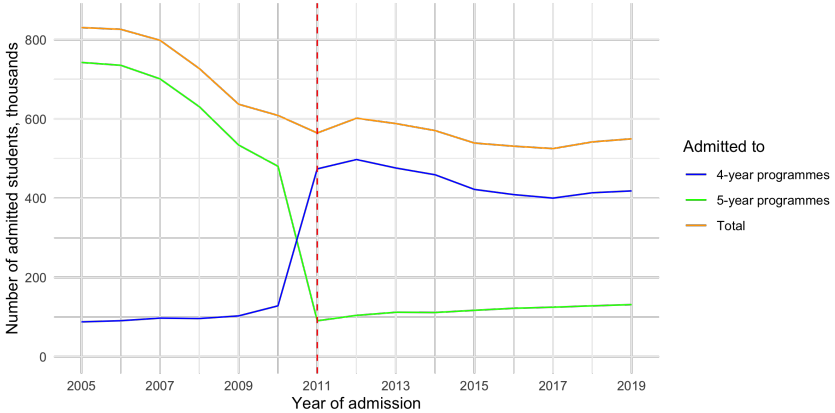
Notes: The red dashed line shows the year of the implementation of the reform.

Figure: The number of 4– and 5–year full–time admitted students in public and private HEIs in 2005–2019, thousands



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Notes: The red dashed line shows the year of the implementation of the reform.

Table: The characteristics of university students by the year of graduation

Variable	2010	2011	2012	2013	2014	2015
Male	0.44	0.44	0.45	0.42	0.40	0.39
	(0.50)	(0.50)	(0.50)	(0.49)	(0.49)	(0.49)
Public institution	0.94	0.96	0.95	0.95	0.96	0.95
	(0.23)	(0.20)	(0.22)	(0.22)	(0.19)	(0.21)
State–funded	0.59	0.60	0.59	0.59	0.59	0.58
	(0.49)	(0.49)	(0.49)	(0.49)	(0.49)	(0.49)

Notes: This table shows sample means and standard deviations from the the National Survey of Graduate Employment carried out by the Russian Federal State Statistics Service.

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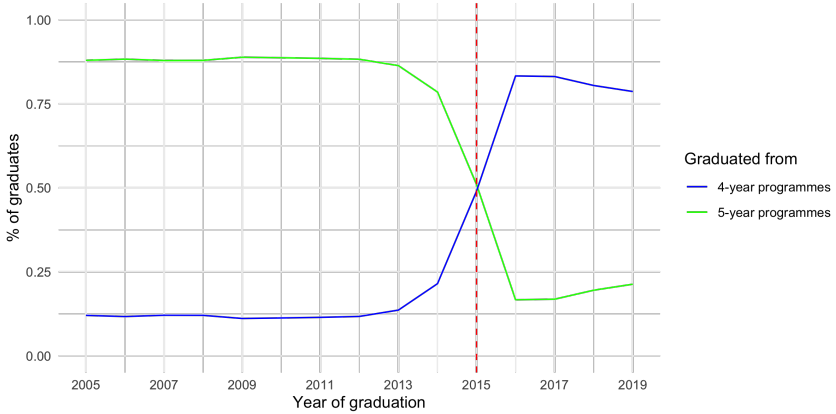
Figure: The share of 4– and 5–year full–time graduates in public and private universities in 2005–2019



Notes: The red dashed line shows the year of a double flow of graduates.

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Table: The characteristics of graduates when they were admitted to university

Variable	2010	2011	2012	2013	2014	2015
Age at the year of application	17.38	17.45	17.62	17.66	17.82	17.73
	(1.09)	(1.21)	(1.26)	(1.27)	(1.39)	(1.35)
Rural residence	0.21	0.23	0.24	0.28	0.27	0.28
	(0.41)	(0.42)	(0.43)	(0.45)	(0.44)	(0.45)
Moscow and Saint Petersburg	0.13	0.13	0.13	0.12	0.10	0.12
	(0.34)	(0.33)	(0.34)	(0.32)	(0.30)	(0.33)

Notes: This table shows sample means and standard deviations from the the National Survey of Graduate Employment carried out by the Russian Federal State Statistics Service.

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